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FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

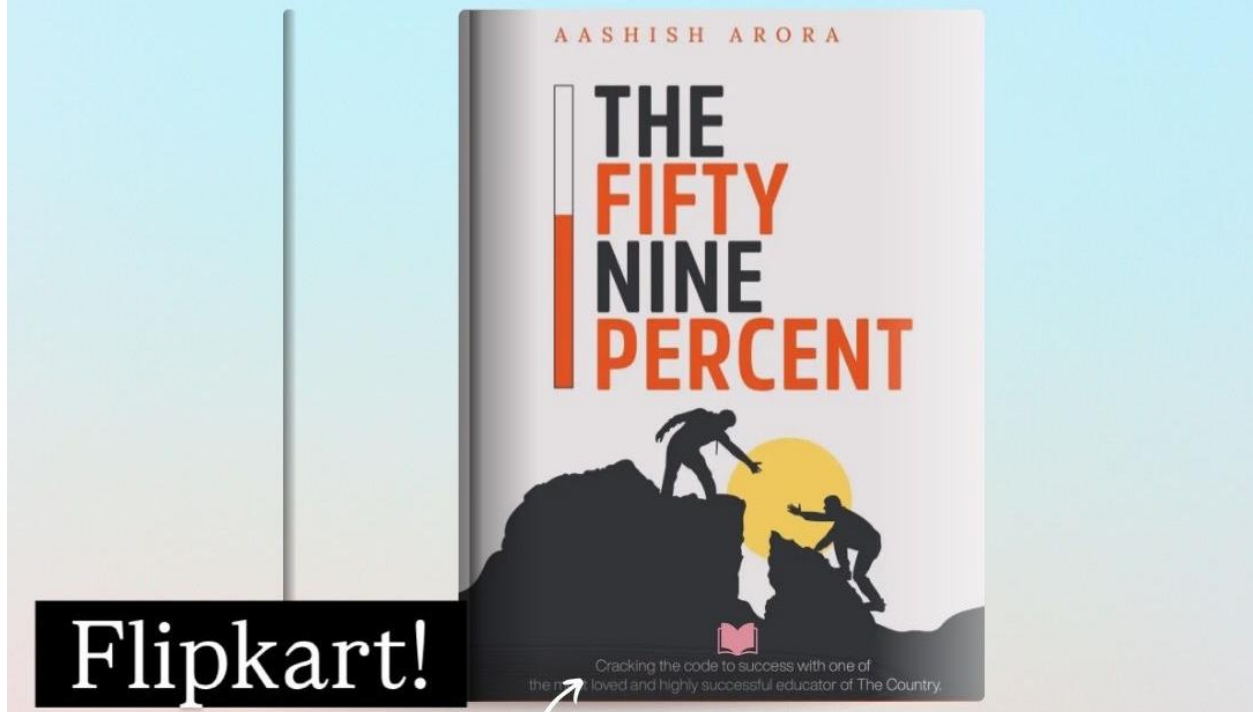


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Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

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1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $46.15\% \text{ of } 1690 + \frac{7}{11} \text{ of } 990 - \sqrt{1600} = ? \times 10$

- A. 131
- B. 137
- C. 143
- D. 149
- E. None of these

2. $75\% \text{ of } 960 - ?\% \text{ of } 800 = \frac{13}{15} \text{ of } 600$

- A. 20
- B. 23
- C. 25
- D. 28
- E. None of these

3. $\sqrt{5184} + \sqrt[3]{46656} + 40\% \text{ of } 625 = ? + 12^2$

- A. 214
- B. 222
- C. 206

D. 230

- E. None of these

4. $(\frac{7}{15} \text{ of } 450 + \frac{3}{8} \text{ of } 640) \div ? = 25$

- A. 12
- B. 15
- C. 21
- D. 18
- E. None of these

5. $6\frac{5}{8} + 4\frac{3}{4} - 2\frac{7}{8} + 3\frac{1}{2} = ?$

- A. $11\frac{1}{2}$
- B. 12
- C. $12\frac{1}{2}$
- D. 13
- E. None of these

6. $[(12^2 + 20^2) \div 8] + 31.25\% \text{ of } 608 = ? + 75$

- A. 183
- B. 177
- C. 189

D. 195

E. None of these

7. 56.25% of $576 + \frac{5}{8}$ of $200 - 7^2 = ? \times 20$

A. 16

B. 18

C. 22

D. 20

E. None of these

8. $20 \times 25 + 75\%$ of $320 = ? \times 16 - \sqrt[3]{21952}$

A. 42

B. 45

C. 48

D. 52

E. None of these

9. 42.85% of $? + 18 \times 32 = \frac{7}{9}$ of $918 + 72$

A. 455

B. 490

C. 525

D. 560

E. None of these

10. $24.75 \times 40 + 18.5 \times 36 - 12.25 \times 32 = ? \times 8$

A. 146

B. 152

C. 164

D. 158

E. None of these

11. $\sqrt{?} + \sqrt[3]{29791} = 75\%$ of $96 + \sqrt{9}$

A. 1764

B. 1849

C. 1936

D. 2025

E. None of these

12. $\frac{11}{13}$ of $845 - \frac{5}{17}$ of $1020 + \frac{8}{9}$ of $621 = ? + 280$

A. 687

B. 667

C. 707

D. 727

E. None of these

13. $54 \times 37 + 46 \times 37 - ? \times 25 = 1850$

A. 69

B. 74

C. 79

D. 84

E. None of these

14. 64% of 25% of 2500 + 12.5% of 704 = ? + 86

A. 386

B. 394

C. 410

D. 402

E. None of these

15. ? $\div$ 8 + 68.75% of 256 = 5^3 + 91

A. 304

B. 312

C. 320

D. 328

E. None of these

16. $8\frac{1}{2} - 3\frac{3}{4} + 5\frac{5}{6} - 2\frac{1}{3} = ?$

A. $8\frac{1}{4}$ B. $7\frac{3}{4}$ C. $8\frac{1}{2}$ D. $9\frac{1}{4}$

E. None of these

17. $\sqrt[3]{17576} + 24^2 - 37.5\% \text{ of } 880 = ? \times 8$

A. 30

B. 34

C. 38

D. 42

E. None of these

18. $\frac{9}{14} \text{ of } 798 + \frac{4}{15} \text{ of } 735 - \sqrt{625} = ? \times 18$

A. 34

B. 36

C. 38

D. 42

E. None of these

19. $[30\% \text{ of } 840 + \frac{7}{8} \text{ of } 512 - (12 \times 13)] \div ? = 34$

A. 12

B. 14

C. 18

D. 16

E. None of these

20. 12.5% of 960 + ?% of 750 = 46.66% of 900

A. 40

B. 38

C. 42

D. 44

E. None of these

Answers:

1. B

2. C

3. A

4. D

5. B

6. A

7. D

8. C

9. B

10. D

11. C

12. A

13. B

14. D

15. C

16. A

17. B

18. C

19. D

20. A

Solutions:

$$1. 46.15\% \text{ of } 1690 + \frac{7}{11} \text{ of } 990 - \sqrt{1600} \\ = ? \times 10$$

$$46.15\% = \frac{6}{13} \text{ and } \sqrt{1600} = 40$$

$$\frac{6}{13} \text{ of } 1690 = 780 \text{ and } \frac{7}{11} \text{ of } 990 = 630$$

$$\text{So, } 780 + 630 - 40 = ? \times 10$$

$$1370 = ? \times 10$$

$$? = 1370 \div 10 = 137$$

$$2. 75\% \text{ of } 960 - ?\% \text{ of } 800 = \frac{13}{15} \text{ of } 600$$

$$75\% \text{ of } 960 = 720 \text{ and } \frac{13}{15} \text{ of } 600 = 520$$

$$\text{So, } 720 - ?\% \text{ of } 800 = 520$$

$$?\% \text{ of } 800 = 720 - 520 = 200$$

$$? = 200 \times 100 \div 800 = 25$$

$$3. \sqrt{5184} + \sqrt[3]{46656} + 40\% \text{ of } 625 = ? + 12^2$$

$$\sqrt{5184} = 72, \sqrt[3]{46656} = 36, \text{ and } 40\% \text{ of } 625 = 250$$

$$\text{So, } 72 + 36 + 250 = ? + 144$$

$$358 = ? + 144$$

$$? = 358 - 144 = 214$$

$$4. \left(\frac{7}{15} \text{ of } 450 + \frac{3}{8} \text{ of } 640 \right) \div ? = 25$$

$$\frac{7}{15} \text{ of } 450 = 210 \text{ and } \frac{3}{8} \text{ of } 640 = 240$$

$$\text{So, } (210 + 240) \div ? = 25$$

$$450 \div ? = 25$$

$$? = 450 \div 25 = 18$$

$$5. 6\frac{5}{8} + 4\frac{3}{4} - 2\frac{7}{8} + 3\frac{1}{2} = ?$$

$$\text{Whole numbers} = 6 + 4 - 2 + 3 = 11$$

$$\text{Fractions} = \frac{5}{8} + \frac{3}{4} - \frac{7}{8} + \frac{1}{2}$$

$$= \frac{5}{8} + \frac{6}{8} - \frac{7}{8} + \frac{4}{8}$$

$$= \frac{8}{8} = 1$$

$$\text{So, } ? = 11 + 1 = 12$$

$$6. [(12^2 + 20^2) \div 8] + 31.25\% \text{ of } 608 = ? + 75$$

$$12^2 = 144, 20^2 = 400, \text{ and } 31.25\% = \frac{5}{16}$$

$$(144 + 400) \div 8 = 544 \div 8 = 68$$

$$31.25\% \text{ of } 608 = 190$$

$$\text{So, } 68 + 190 = ? + 75$$

$$258 = ? + 75$$

$$? = 258 - 75 = 183$$

$$7. 56.25\% \text{ of } 576 + \frac{5}{8} \text{ of } 200 - 7^2 = ? \times 20$$

$$56.25\% = \frac{9}{16} \text{ and } 7^2 = 49$$

$$56.25\% \text{ of } 576 = 324 \text{ and } \frac{5}{8} \text{ of } 200 = 125$$

$$\text{So, } 324 + 125 - 49 = ? \times 20$$

$$400 = ? \times 20$$

$$? = 400 \div 20 = 20$$

$$8. 20 \times 25 + 75\% \text{ of } 320 = ? \times 16 - \sqrt[3]{21952}$$

$$20 \times 25 = 500, 75\% \text{ of } 320 = 240, \text{ and } \sqrt[3]{21952} = 28$$

$$\text{So, } 500 + 240 = ? \times 16 - 28$$

$$740 + 28 = ? \times 16$$

$$768 = ? \times 16$$

$$? = 768 \div 16 = 48$$

$$9. 42.85\% \text{ of } ? + 18 \times 32 = \frac{7}{9} \text{ of } 918 + 72$$

$$42.85\% = \frac{3}{7}, 18 \times 32 = 576, \text{ and } \frac{7}{9} \text{ of } 918 = 714$$

$$\text{So, } \frac{3}{7} \text{ of } ? + 576 = 714 + 72$$

$$\frac{3}{7} \text{ of } ? + 576 = 786$$

$$\frac{3}{7} \text{ of } ? = 210$$

$$? = 210 \times 7 \div 3 = 490$$

$$10. 24.75 \times 40 + 18.5 \times 36 - 12.25 \times 32 = ? \times 8$$

$$24.75 \times 40 = 990, 18.5 \times 36 = 666, \text{ and } 12.25 \times 32 = 392$$

$$\text{So, } 990 + 666 - 392 = ? \times 8$$

$$1264 = ? \times 8$$

$$? = 1264 \div 8 = 158$$

$$11. \sqrt{?} + \sqrt[3]{29791} = 75\% \text{ of } 96 + \sqrt{9}$$

$$\sqrt[3]{29791} = 31, 75\% \text{ of } 96 = 72, \text{ and } \sqrt{9} = 3$$

$$\text{So, } \sqrt{?} + 31 = 72 + 3$$

$$\sqrt{?} + 31 = 75$$

$$\sqrt{?} = 44$$

$$? = 44^2 = 1936$$

$$12. \frac{11}{13} \text{ of } 845 - \frac{5}{17} \text{ of } 1020 + \frac{8}{9} \text{ of } 621 = ? + 280$$

$$\frac{11}{13} \text{ of } 845 = 715, \frac{5}{17} \text{ of } 1020 = 300, \text{ and } \frac{8}{9} \text{ of } 621 = 552$$

$$\text{So, } 715 - 300 + 552 = ? + 280$$

$$967 = ? + 280$$

$$? = 967 - 280 = 687$$

$$13. 54 \times 37 + 46 \times 37 - ? \times 25 = 1850$$

$$54 \times 37 + 46 \times 37 = 37 \times (54 + 46)$$

$$= 37 \times 100 = 3700$$

$$\text{So, } 3700 - ? \times 25 = 1850$$

$$? \times 25 = 3700 - 1850 = 1850$$

$$? = 1850 \div 25 = 74$$

$$14. 64\% \text{ of } 25\% \text{ of } 2500 + 12.5\% \text{ of } 704 = ? + 86$$

$$25\% \text{ of } 2500 = 625, 64\% \text{ of } 625 = 400, \text{ and } 12.5\% = \frac{1}{8}$$

$$12.5\% \text{ of } 704 = 88$$

$$\text{So, } 400 + 88 = ? + 86$$

$$488 = ? + 86$$

$$? = 488 - 86 = 402$$

$$15. ? \div 8 + 68.75\% \text{ of } 256 = 5^3 + 91$$

$$68.75\% = \frac{11}{16} \text{ and } 5^3 = 125$$

$$68.75\% \text{ of } 256 = 176$$

$$\text{So, } ? \div 8 + 176 = 125 + 91$$

$$? \div 8 + 176 = 216$$

$$? \div 8 = 40$$

$$? = 40 \times 8 = 320$$

$$16. 8\frac{1}{2} - 3\frac{3}{4} + 5\frac{5}{6} - 2\frac{1}{3} = ?$$

$$\text{Whole numbers} = 8 - 3 + 5 - 2 = 8$$

$$\text{Fractions} = \frac{1}{2} - \frac{3}{4} + \frac{5}{6} - \frac{1}{3}$$

$$= \frac{6}{12} - \frac{9}{12} + \frac{10}{12} - \frac{4}{12}$$

$$= \frac{3}{12} = \frac{1}{4}$$

$$\text{So, } ? = 8\frac{1}{4}$$

$$17. \sqrt[3]{17576} + 24^2 - 37.5\% \text{ of } 880 = ? \times 8$$

$$\sqrt[3]{17576} = 26, 24^2 = 576, \text{ and } 37.5\% = \frac{3}{8}$$

$$37.5\% \text{ of } 880 = 330$$

$$\text{So, } 26 + 576 - 330 = ? \times 8$$

$$272 = ? \times 8$$

$$? = 272 \div 8 = 34$$

$$18. \frac{9}{14} \text{ of } 798 + \frac{4}{15} \text{ of } 735 - \sqrt{625} = ? \times 18$$

$$\frac{9}{14} \text{ of } 798 = 513, \frac{4}{15} \text{ of } 735 = 196, \text{ and } \sqrt{625} = 25$$

$$\text{So, } 513 + 196 - 25 = ? \times 18$$

$$684 = ? \times 18$$

$$? = 684 \div 18 = 38$$

$$19. [30\% \text{ of } 840 + \frac{7}{8} \text{ of } 512 - (12 \times 13)] \div ? = 34$$

$$30\% \text{ of } 840 = 252, \frac{7}{8} \text{ of } 512 = 448, \text{ and}$$

$$12 \times 13 = 156$$

$$\text{So, } [252 + 448 - 156] \div ? = 34$$

$$544 \div ? = 34$$

$$? = 544 \div 34 = 16$$

$$20. 12.5\% \text{ of } 960 + ?\% \text{ of } 750 = 46.66\% \text{ of } 900$$

$$12.5\% = \frac{1}{8} \text{ and } 46.66\% = \frac{7}{15}$$

$$12.5\% \text{ of } 960 = 120 \text{ and } 46.66\% \text{ of } 900 = 420$$

$$\text{So, } 120 + ?\% \text{ of } 750 = 420$$

$$?\% \text{ of } 750 = 300$$

$$? = 300 \times 100 \div 750 = 40$$

2. ARITHMETIC QUESTIONS

Q1: Mehul invested Rs. 25000 in Scheme A, which gives simple interest at $r\%$ p.a. for 2 years. The interest earned from Scheme A was then invested in Scheme B, which gives compound interest at 5% p.a. for 2 years. If the amount received from Scheme B is Rs. 8820, find the value of r .

मेहुल ने Rs. 25000 योजना A में निवेश किए, जिसमें 2 वर्षों के लिए $r\%$ वार्षिक दर से साधारण ब्याज मिलता है। योजना A से प्राप्त ब्याज को फिर योजना B में निवेश किया गया, जिसमें 2 वर्षों के लिए 5% वार्षिक दर से चक्रवृद्धि ब्याज मिलता है। यदि योजना B से प्राप्त राशि Rs. 8820 है, तो r का मान ज्ञात कीजिए।

- (A) 12
- (B) 14
- (C) 18
- (D) 16
- (E) 20

Q2: Two cars start at the same time from cities M and N and move toward each other. The distance between the two cities is 360 km. The first car can cover the whole distance in 6 hours, while the second car can cover the same distance in 12 hours. After how many hours from the start will they meet?

दो कारें शहर M और शहर N से एक ही समय पर एक-दूसरे की ओर चलना शुरू करती हैं। दोनों शहरों के बीच की दूरी 360 किमी है। पहली कार पूरी दूरी 6 घंटे में तय कर सकती है, जबकि दूसरी कार वही दूरी 12 घंटे में तय कर सकती है। वे शुरू होने के कितने घंटे बाद मिलेंगी?

- (A) 3 hours
- (B) 4 hours
- (C) 5 hours
- (D) 6 hours
- (E) 8 hours

Q3: A boat covers 75 km downstream in t hours and 81 km upstream in $(t + 4)$ hours. The ratio of the speed of the boat in still water to its downstream speed is $4:5$. Find the time taken by the boat to cover 45 km downstream and 45 km upstream.

एक नाव 75 किमी धारा के अनुकूल दूरी t घंटे में और 81 किमी धारा के प्रतिकूल दूरी $(t + 4)$

घंटे में तय करती है। शांत जल में नाव की गति और धारा के अनुकूल गति का अनुपात 4:5 है। नाव को 45 किमी धारा के अनुकूल और 45 किमी धारा के प्रतिकूल दूरी तय करने में कितना समय लगेगा?

- (A) 6 hours
- (B) 7 hours
- (C) 8 hours
- (D) 9 hours
- (E) 10 hours

Q4: A milk vendor mixed $(x + 30)$ liters of water with 240 liters of pure milk. He then replaced 60 liters of the mixture with 60 liters of pure milk. If the quantity of milk in the final mixture becomes 260 liters, find x .

एक दूध विक्रेता ने 240 लीटर शुद्ध दूध में $(x + 30)$ लीटर पानी मिलाया। इसके बाद उसने मिश्रण के 60 लीटर को 60 लीटर शुद्ध दूध से बदल दिया। यदि अंतिम मिश्रण में दूध की मात्रा 260 लीटर हो जाती है, तो x ज्ञात कीजिए।

- (A) 90
- (B) 75
- (C) 105
- (D) 120
- (E) 150

Q5: Vivek bought a laptop for Rs. A and sold it for Rs. B, earning a profit of 30%. If he had bought it at 10% less and sold it for Rs. 220 less, he would have earned a profit of 20%. Find the value of $B - A$.

विवेक ने एक लैपटॉप Rs. A में खरीदा और Rs. B में बेचकर 30% लाभ कमाया। यदि उसने इसे 10% कम कीमत पर खरीदा होता और Rs. 220 कम में बेचा होता, तो उसे 20% लाभ होता। $B - A$ का मान ज्ञात कीजिए।

- (A) Rs. 240
- (B) Rs. 360
- (C) Rs. 250
- (D) Rs. 450
- (E) Rs. 300

Q6: A trader marked an article $3K\%$ above its cost price. He allowed a discount of 20% and sold it at a loss of 8% for Rs 736. Find the value of K .

एक व्यापारी ने एक वस्तु को उसके क्रय मूल्य से $3K\%$ अधिक अंकित किया। उसने 20% की छूट दी और वस्तु को 8% हानि पर रु. 736 में बेच दिया। K का मान ज्ञात कीजिए।

- (A) 4
- (B) 6
- (C) 5
- (D) 8
- (E) 10

Q7: The sum of the length of a rectangle and the side of a square is 58 meters. The perimeter of the rectangle is 76 meters, and its breadth is 14 meters. Find the side of the square.

एक आयत की लंबाई और एक वर्ग की भुजा का योग 58 मीटर है। आयत का परिमाप 76 मीटर है और उसकी चौड़ाई 14 मीटर है। वर्ग की भुजा ज्ञात कीजिए।

- (A) 30 meters
- (B) 36 meters
- (C) 42 meters
- (D) 28 meters
- (E) 34 meters

Q8: Train M runs at a speed of 54 km/h and crosses a platform whose length is one-fourth of its own length in 50 seconds. Find the time taken by Train M to cross Train N, whose length is twice the length of Train M and which runs at 126 km/h in the opposite direction.

ट्रेन M 54 किमी/घंटा की गति से चलती है और अपनी लंबाई के एक-चौथाई लंबे प्लेटफॉर्म को 50 सेकंड में पार करती है। ट्रेन M को ट्रेन N को पार करने में कितना समय लगेगा, जिसकी लंबाई ट्रेन M की लंबाई की दोगुनी है और जो विपरीत दिशा में 126 किमी/घंटा की गति से चलती है?

- (A) 32 seconds
- (B) 36 seconds
- (C) 40 seconds
- (D) 45 seconds
- (E) 48 seconds

Q9: A trader bought a chair for Rs M and sold it for Rs N at a profit of 25%. If he had bought the chair at 20% less and sold it for Rs 132 less, he would have earned a profit of 15%. Find the value of $N - M$.

एक व्यापारी ने एक कुर्सी रु. M में खरीदी और उसे 25% लाभ पर रु. N में बेच दिया। यदि उसने कुर्सी को 20% कम कीमत पर खरीदा होता और रु. 132 कम में बेचा होता, तो उसे 15% लाभ होता। $N - M$ का मान ज्ञात कीजिए।

- (A) Rs 80
- (B) Rs 120
- (C) Rs 90
- (D) Rs 100
- (E) Rs 150

Q10: A boat covers 120 km downstream in m hours and 144 km upstream in $(m + 4)$ hours. The ratio of the speed of the boat in still water to its downstream speed is 5:6. Find the time taken by the boat to cover 48 km downstream and 48 km upstream.

एक नाव धारा के अनुकूल 120 किमी की दूरी m घंटों में और धारा के प्रतिकूल 144 किमी की दूरी $(m + 4)$ घंटों में तय करती है। शांत जल में नाव की गति और उसकी धारा के अनुकूल गति का अनुपात 5:6 है। नाव को 48 किमी धारा के अनुकूल और 48 किमी धारा के प्रतिकूल दूरी तय करने में कितना समय लगेगा?

- (A) 6 hours
- (B) 5 hours
- (C) 8 hours
- (D) 7 hours
- (E) 4 hours

Q11: Pipe P can fill 25% of a tank in 5 hours, and pipe Q can fill 30% of the tank in 6 hours. Pipe R is an outlet pipe and can empty the full tank in 50 hours. If the tank is already 60% filled, find the time taken by all three pipes together to fill the remaining part of the tank.

पाइप P एक टंकी का 25% भाग 5 घंटे में भर सकता है और पाइप Q टंकी का 30% भाग 6 घंटे में भर सकता है। पाइप R एक निकासी पाइप है और पूरी टंकी को 50 घंटे में खाली कर सकता है। यदि टंकी पहले से 60% भरी हुई है, तो तीनों पाइप मिलकर शेष भाग को भरने में कितना समय लेंगे?

- (A) 4 hours
- (B) 6 hours
- (C) 5 hours
- (D) 8 hours
- (E) 10 hours

Q12: The sum of the breadth of a rectangle and the side of a square is 34 meters. If the perimeter of the rectangle is 72 meters and the length of the rectangle is 22 meters, find the side of the square.

एक आयत की चौड़ाई और एक वर्ग की भुजा का योग 34 मीटर है। यदि आयत का परिमाण 72 मीटर है और आयत की लंबाई 22 मीटर है, तो वर्ग की भुजा ज्ञात कीजिए।

- (A) 16 meters
- (B) 18 meters
- (C) 24 meters
- (D) 22 meters
- (E) 20 meters

Q13: Aditya spends 35% of his monthly salary on rent and 25% on food. From the remaining amount, he gives money to his father and mother in the ratio of 7 : 5. If the difference between the amounts given to his father and mother is Rs. 8000, find Aditya's monthly salary.

आदित्य अपने मासिक वेतन का 35% किराए पर और 25% भोजन पर खर्च करता है। शेष राशि में से वह अपने पिता और माता को 7 : 5 के अनुपात में पैसे देता है। यदि पिता और माता को दी गई राशि का अंतर Rs. 8000 है, तो आदित्य का मासिक वेतन ज्ञात कीजिए।

- (A) Rs. 100000
- (B) Rs. 120000
- (C) Rs. 90000
- (D) Rs. 150000
- (E) Rs. 80000

Q14: Train X, running at 108 km/hr, crosses a platform whose length is $\frac{2}{3}$ of its own length in 40 seconds. Find the time taken by Train X to cross Train Y, whose length is $\frac{4}{3}$ times the length of Train X and which is running at 72 km/hr in the same direction.

ट्रेन X, 108 किमी/घंटा की गति से चलते हुए, एक ऐसे प्लेटफॉर्म को 40 सेकंड में पार करती है जिसकी लंबाई ट्रेन X की अपनी लंबाई की $\frac{2}{3}$ है। ट्रेन X को ट्रेन Y को पार करने में कितना समय लगेगा, जिसकी लंबाई ट्रेन X की लंबाई की $\frac{4}{3}$ गुनी है और जो उसी दिशा में 72 किमी/घंटा की गति से चल रही है?

- (A) 168 seconds
- (B) 144 seconds
- (C) 180 seconds
- (D) 156 seconds
- (E) 192 seconds

Q15: Bhavna started a business by investing Rs. 150000. After 4 months, Charu joined her with Rs. 210000. After another 4 months, Bhavna withdrew $\frac{1}{5}$ of her

investment, and Charu added Rs. 30000 more. At the end of the year, the total profit was Rs. 87000. Find Bhavna's profit share.

भावना ने Rs. 150000 निवेश करके एक व्यवसाय शुरू किया। 4 महीने बाद, चारु Rs. 210000 के साथ व्यवसाय में शामिल हुई। अगले 4 महीने बाद, भावना ने अपने निवेश का $\frac{1}{5}$ भाग निकाल लिया और चारु ने Rs. 30000 और जोड़ दिए। वर्ष के अंत में कुल लाभ Rs. 87000 था। भावना का लाभांश ज्ञात कीजिए।

- (A) Rs. 39000
- (B) Rs. 45000
- (C) Rs. 48000
- (D) Rs. 42000
- (E) Rs. 36000

Q16: Ishaan and Lakshya invested in a business in the ratio 3 : 5 respectively. Lakshya invested for 2 months less than the time for which Ishaan invested. If Ishaan's profit is 10% less than Lakshya's profit, find the time for which Lakshya invested.

ईशान और लक्ष्य ने एक व्यवसाय में क्रमशः 3 : 5 के अनुपात में निवेश किया। लक्ष्य ने ईशान की तुलना में 2 महीने कम समय के लिए निवेश किया। यदि ईशान का लाभ, लक्ष्य के लाभ से 10% कम है, तो लक्ष्य ने कितने समय के लिए निवेश किया?

- (A) 5 months
- (B) 7 months
- (C) 4 months
- (D) 6 months
- (E) 8 months

Q17: A dairy owner mixed $(z + 40)$ liters of water in 200 liters of milk. Then he replaced 60 liters of the mixture with milk. If the quantity of milk in the final mixture is 220 liters, find z .

एक डेयरी मालिक ने 200 लीटर दूध में $(z + 40)$ लीटर पानी मिलाया। फिर उसने मिश्रण के 60 लीटर को दूध से बदल दिया। यदि अंतिम मिश्रण में दूध की मात्रा 220 लीटर है, तो z ज्ञात कीजिए।

- (A) 45 liters
- (B) 55 liters
- (C) 70 liters
- (D) 50 liters
- (E) 60 liters

Q18: Nisha spends 25% of her monthly salary on house rent and 35% of her monthly salary on other expenses. She divides the remaining amount completely between her sister and brother in the ratio 5 : 3. If the difference between the amounts given to her sister and brother is Rs. 6000, find Nisha's monthly salary.

निशा अपनी मासिक आय का 25% घर के किराए पर और 35% अन्य खर्चों पर खर्च करती है। वह बची हुई पूरी राशि अपनी बहन और भाई के बीच 5 : 3 के अनुपात में बांटती है। यदि बहन और भाई को दी गई राशि के बीच अंतर Rs. 6000 है, तो निशा की मासिक आय ज्ञात कीजिए।

- (A) Rs. 60000
- (B) Rs. 52000
- (C) Rs. 64000
- (D) Rs. 58000
- (E) Rs. 72000

Q19: Two buses start at the same time from two cities towards each other. The distance between the cities is 720 km. The first bus can cover the whole distance in 6 hours, and the second bus can cover the same distance in 12 hours. After how much time will they meet?

दो बसें दो शहरों से एक ही समय पर एक-दूसरे की ओर चलना शुरू करती हैं। दोनों शहरों के बीच की दूरी 720 किमी है। पहली बस पूरी दूरी 6 घंटे में तय कर सकती है और दूसरी बस वही दूरी 12 घंटे में तय कर सकती है। वे कितने समय बाद मिलेंगी?

- (A) 5 hours
- (B) 6 hours
- (C) 3 hours
- (D) 4 hours
- (E) 8 hours

Q20: Pipe P can fill 20% of a tank in 4 hours, and Pipe Q can fill 30% of the same tank in 6 hours. Pipe R is an outlet pipe and can empty the completely filled tank in 30 hours. If the tank is already 40% filled, find the time taken by all three pipes together to fill the remaining tank.

पाइप P किसी टैंक का 20% भाग 4 घंटे में भर सकता है और पाइप Q उसी टैंक का 30% भाग 6 घंटे में भर सकता है। पाइप R एक निकासी पाइप है और भरे हुए टैंक को 30 घंटे में खाली कर सकता है। यदि टैंक पहले से 40% भरा हुआ है, तो तीनों पाइप मिलकर शेष टैंक को कितने समय में भरेंगे?

- (A) 8 hours
- (B) 9 hours

- (C) 10 hours
(D) 12 hours
(E) 6 hours

Answers:

- | | |
|------|------|
| 1. D | 11.C |
| 2. B | 12.E |
| 3. C | 13.B |
| 4. A | 14.A |
| 5. E | 15.D |
| 6. C | 16.C |
| 7. E | 17.E |
| 8. B | 18.A |
| 9. D | 19.D |
| 10.B | 20.B |

Solutions:**1. Option D**

Amount received from Scheme B = Rs. 8820

Rate of compound interest in Scheme B = 5% p.a.

Time = 2 years

Let the interest earned from Scheme A be Rs. I.

This amount I was invested in Scheme B.

Amount in Scheme B = $I \times \frac{105}{100} \times \frac{105}{100}$

$8820 = I \times \frac{105}{100} \times \frac{105}{100}$

$I = \frac{8820 \times 100 \times 100}{(105 \times 105)}$

$I = 8000$

So, interest earned from Scheme A = Rs. 8000

Now, principal in Scheme A = Rs. 25000

Rate = r% p.a.

Time = 2 years

Simple interest = $\frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$

$8000 = \frac{25000 \times r \times 2}{100}$

$$8000 = 500r$$

$$r = 8000/500$$

$$r = 16$$

2. Option B

Distance between the two cities = 360 km

The first car covers 360 km in 6 hours.

Speed of the first car = $360/6 = 60$ km/hr

The second car covers 360 km in 12 hours.

Speed of the second car = $360/12 = 30$ km/hr

Since both cars are moving toward each other, their relative speed will be the sum of their speeds.

Relative speed = $60 + 30 = 90$ km/hr

Time taken to meet = Distance/Relative speed

Time taken = $360/90$

Time taken = 4 hours

Therefore, both cars will meet after 4 hours.

3. Option C

Ratio of speed in still water to downstream speed = 4:5

Let speed in still water = $4a$

Let downstream speed = $5a$

Speed of stream = Downstream speed – Speed in still water

Speed of stream = $5a - 4a = a$

Upstream speed = Speed in still water – Speed of stream

Upstream speed = $4a - a = 3a$

The boat covers 75 km downstream in t hours.

So, $t = 75/5a = 15/a$

The boat covers 81 km upstream in $(t + 4)$ hours.

So, $t + 4 = 81/3a = 27/a$

Now,

$t + 4 - t = 27/a - 15/a$

$4 = 12/a$

$4a = 12$

$a = 3$

Downstream speed = $5a = 5 \times 3 = 15$ km/hr

Upstream speed = $3a = 3 \times 3 = 9$ km/hr

Time taken to cover 45 km downstream = $45/15 = 3$ hours

Time taken to cover 45 km upstream = $45/9 = 5$ hours

Total time = $3 + 5 = 8$ hours

4. Option A

Initial quantity of milk = 240 liters

Initial quantity of water = $x + 30$ liters

Total mixture = $240 + x + 30 = x + 270$ liters

Final quantity of milk = 260 liters

Increase in milk quantity = $260 - 240 = 20$ liters

The milk quantity increases because 60 liters of mixture is replaced by 60 liters of pure milk.

So, the increase in milk is equal to the quantity of water removed from the mixture.

Water fraction in the mixture = $(x + 30)/(x + 270)$

Water removed in 60 liters of mixture = $60 \times (x + 30)/(x + 270)$

According to the question:

$$60 \times (x + 30)/(x + 270) = 20$$

$$60(x + 30) = 20(x + 270)$$

$$60x + 1800 = 20x + 5400$$

$$60x - 20x = 5400 - 1800$$

$$40x = 3600$$

$$x = 3600/40$$

$$x = 90$$

5. Option E

Vivek bought the laptop for Rs. A.

He sold it for Rs. B and earned 30% profit.

So, $B = 130\%$ of A

$$B = 130A/100$$

In the second case, he bought it at 10% less.

New cost price = 90% of A

$$\text{New cost price} = 90A/100$$

He sold it for Rs. 220 less than B.

$$\text{New selling price} = B - 220$$

He earned 20% profit in the second case.

So, new selling price = 120% of new cost price

$$B - 220 = 120/100 \times 90A/100$$

$$B - 220 = 108A/100$$

Now put $B = 130A/100$.

$$130A/100 - 220 = 108A/100$$

$$130A/100 - 108A/100 = 220$$

$$22A/100 = 220$$

$$A = 220 \times 100/22$$

$$A = 1000$$

Now, $B - A$ is the profit in the first case.

Profit = 30% of A

$$B - A = 30 \times 1000/100$$

$$B - A = 300$$

Therefore, the value of $B - A$ is Rs. 300.

6. Option C

Selling price of the article = Rs 736.

The article is sold at a loss of 8%.

So, selling price = 92% of cost price.

$$\text{Cost price} = 736 \times 100/92$$

$$\text{Cost price} = \text{Rs } 800$$

The trader allowed a discount of 20%.

So, selling price = 80% of marked price.

$$\text{Marked price} = 736 \times 100/80$$

$$\text{Marked price} = \text{Rs } 920$$

$$\text{Increase over cost price} = 920 - 800 = \text{Rs } 120$$

$$\text{Markup percentage} = 120 \times 100/800$$

$$\text{Markup percentage} = 15\%$$

According to the question, markup percentage = 3K%.

$$\text{So, } 3K = 15$$

$$K = 15/3$$

$$K = 5$$

7. Option E

Perimeter of the rectangle = 76 meters.

Breadth of the rectangle = 14 meters.

Formula for perimeter of rectangle:

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$$

$$\text{So, } 76 = 2 \times (\text{Length} + 14)$$

$$\text{Length} + 14 = 76/2$$

$$\text{Length} + 14 = 38$$

$$\text{Length} = 38 - 14$$

$$\text{Length} = 24 \text{ meters}$$

Given, length of rectangle + side of square = 58 meters.

$$\text{So, } 24 + \text{side of square} = 58$$

$$\text{Side of square} = 58 - 24$$

$$\text{Side of square} = 34 \text{ meters}$$

8. Option B

Speed of Train M = 54 km/h.

Convert km/h into m/s:

$$54 \times 5/18 = 15 \text{ m/s}$$

Let the length of Train M be L meters.

$$\text{Length of platform} = L/4.$$

When Train M crosses the platform, total distance covered:

$$L + L/4 = 5L/4$$

Train M crosses it in 50 seconds.

$$\text{Distance covered in 50 seconds} = 15 \times 50 = 750 \text{ meters.}$$

$$\text{So, } 5L/4 = 750$$

$$L = 750 \times 4/5$$

$$L = 600 \text{ meters}$$

Length of Train M = 600 meters.

$$\text{Length of Train N} = 2 \times 600 = 1200 \text{ meters.}$$

Speed of Train N = 126 km/h.

Convert into m/s:

$$126 \times 5/18 = 35 \text{ m/s}$$

Both trains are running in opposite directions.

$$\text{Relative speed} = 15 + 35 = 50 \text{ m/s}$$

Total distance to be covered while crossing:

$$600 + 1200 = 1800 \text{ meters}$$

Time taken = Total distance/Relative speed

$$\text{Time taken} = 1800/50$$

$$\text{Time taken} = 36 \text{ seconds}$$

9. Option D

Let the cost price of the chair be Rs M.

Selling price = Rs N.

The chair is sold at a profit of 25%.

$$\text{So, } N = M \times 125/100$$

$$N = 5M/4$$

In the second case, the chair is bought at 20% less.

$$\text{New cost price} = M \times 80/100$$

$$\text{New cost price} = 4M/5$$

The chair is sold for Rs 132 less.

$$\text{New selling price} = N - 132$$

Profit in the second case = 15%.

$$\text{So, new selling price} = \text{New cost price} \times 115/100$$

$$N - 132 = 4M/5 \times 115/100$$

$$N - 132 = 23M/25$$

$$\text{Now put } N = 5M/4.$$

$$5M/4 - 132 = 23M/25$$

Take LCM of 4 and 25 as 100.

$$125M/100 - 23M/25 = 132$$

$$125M/100 - 92M/100 = 132$$

$$33M/100 = 132$$

$$33M = 13200$$

$$M = 13200/33$$

$$M = 400$$

$$\text{Now, } N = 5M/4$$

$$N = 5 \times 400/4$$

$$N = 500$$

$$\text{Required value} = N - M$$

$$N - M = 500 - 400$$

$$N - M = \text{Rs } 100$$

Therefore, the value of $N - M$ is Rs 100.

10. Option B

Let the speed of the boat in still water be $5x$.

Downstream speed = $6x$.

$$\text{So, speed of stream} = 6x - 5x = x.$$

$$\text{Upstream speed} = 5x - x = 4x.$$

$$\text{Therefore, downstream speed : upstream speed} = 6x : 4x = 3 : 2.$$

Downstream distance = 120 km and time = m hours.

Upstream distance = 144 km and time = $(m + 4)$ hours.

So, downstream speed/upstream speed = $3/2$.

Now,

$$(120/m) / (144/(m + 4)) = 3/2$$

$$120 \times (m + 4)/(144m) = 3/2$$

Now cross multiply:

$$240 \times (m + 4) = 432m$$

$$240m + 960 = 432m$$

$$960 = 432m - 240m$$

$$960 = 192m$$

$$m = 960/192$$

$$m = 5$$

$$\text{Downstream speed} = 120/5 = 24 \text{ km/h}$$

$$\text{Upstream time} = m + 4 = 5 + 4 = 9 \text{ hours}$$

$$\text{Upstream speed} = 144/9 = 16 \text{ km/h}$$

$$\text{Time taken to cover 48 km downstream} = 48/24 = 2 \text{ hours}$$

$$\text{Time taken to cover 48 km upstream} = 48/16 = 3 \text{ hours}$$

$$\text{Total time} = 2 + 3 = 5 \text{ hours}$$

11. Option C

Pipe P fills 25% of the tank in 5 hours.

$$25\% = 25/100 = 1/4$$

$$\text{So, work done by pipe P in 1 hour} = 1/4 \times 1/5 = 1/20$$

Pipe Q fills 30% of the tank in 6 hours.

$$30\% = 30/100 = 3/10$$

$$\text{So, work done by pipe Q in 1 hour} = 3/10 \times 1/6 = 3/60 = 1/20$$

Pipe R empties the full tank in 50 hours.

$$\text{So, work done by pipe R in 1 hour} = -1/50$$

$$\text{Net work done by all three pipes in 1 hour} = 1/20 + 1/20 - 1/50$$

$$\text{LCM of 20, 20 and 50} = 100$$

$$\text{Net work} = 5/100 + 5/100 - 2/100 = 8/100 = 2/25$$

The tank is already 60% filled.

$$\text{Remaining part} = 40\% = 40/100 = 2/5$$

$$\text{Required time} = \text{Remaining work}/\text{Net one hour work}$$

$$\text{Required time} = 2/5 \times 25/2 = 5 \text{ hours}$$

12. Option E

Perimeter of rectangle = 72 meters

Formula of perimeter of rectangle = $2 \times (\text{Length} + \text{Breadth})$

So, $2 \times (\text{Length} + \text{Breadth}) = 72$

$\text{Length} + \text{Breadth} = 72/2 = 36$ meters

Length of rectangle = 22 meters

Breadth of rectangle = $36 - 22 = 14$ meters

Given, breadth of rectangle + side of square = 34 meters

So, side of square = $34 - 14 = 20$ meters

13. Option B

Aditya spends 35% on rent and 25% on food.

Total spending = $35\% + 25\% = 60\%$

Remaining amount = $100\% - 60\% = 40\%$ of salary

The remaining amount is divided between father and mother in the ratio of 7 : 5.

Difference in ratio = $7 - 5 = 2$ parts

Given difference = Rs. 8000

So, 2 parts = Rs. 8000

1 part = $8000/2 = \text{Rs. } 4000$

Total ratio parts = $7 + 5 = 12$ parts

Remaining amount = $12 \times 4000 = \text{Rs. } 48000$

This remaining amount is 40% of salary.

So, 40% of salary = Rs. 48000

Salary = $48000 \times 100/40 = \text{Rs. } 120000$

14. Option A

Speed of Train X = 108 km/hr

Convert km/hr into m/sec.

Speed of Train X = $108 \times 5/18 = 30$ m/sec

Let the length of Train X be L meters.

Length of platform = $2/3$ of L = $2L/3$

Distance covered while crossing the platform = Length of train + Length of platform

Distance = $L + 2L/3 = 5L/3$

Train X crosses the platform in 40 seconds.

So, distance covered = Speed $\times$ Time

$5L/3 = 30 \times 40$

$5L/3 = 1200$

$$L = 1200 \times \frac{3}{5} = 720 \text{ meters}$$

So, length of Train X = 720 meters

Length of Train Y = $\frac{4}{3}$ of Train X

$$\text{Length of Train Y} = 720 \times \frac{4}{3} = 960 \text{ meters}$$

Speed of Train Y = 72 km/hr

$$\text{Speed of Train Y} = 72 \times \frac{5}{18} = 20 \text{ m/sec}$$

Both trains are moving in the same direction.

$$\text{Relative speed} = 30 - 20 = 10 \text{ m/sec}$$

Distance to be covered to cross Train Y = Length of Train X + Length of Train Y

$$\text{Distance} = 720 + 960 = 1680 \text{ meters}$$

$$\text{Time taken} = \frac{\text{Distance}}{\text{Relative speed}}$$

$$\text{Time taken} = \frac{1680}{10} = 168 \text{ seconds}$$

15. Option D

Bhavna invested Rs. 150000 initially.

For the first 8 months, Bhavna's investment = Rs. 150000

$$\text{Bhavna's capital-time value for 8 months} = 150000 \times 8 = 1200000$$

After 8 months, Bhavna withdrew $\frac{1}{5}$ of her investment.

$$\text{Amount withdrawn} = 150000 \times \frac{1}{5} = \text{Rs. } 30000$$

$$\text{Remaining investment of Bhavna} = 150000 - 30000 = \text{Rs. } 120000$$

$$\text{For the last 4 months, Bhavna's capital-time value} = 120000 \times 4 = 480000$$

$$\text{Total capital-time value of Bhavna} = 1200000 + 480000 = 1680000$$

Charu joined after 4 months with Rs. 210000.

$$\text{So, for the next 4 months, Charu's capital-time value} = 210000 \times 4 = 840000$$

After that, Charu added Rs. 30000 more.

$$\text{New investment of Charu} = 210000 + 30000 = \text{Rs. } 240000$$

$$\text{For the last 4 months, Charu's capital-time value} = 240000 \times 4 = 960000$$

$$\text{Total capital-time value of Charu} = 840000 + 960000 = 1800000$$

$$\text{Profit ratio of Bhavna and Charu} = 1680000 : 1800000$$

Divide both terms by 120000.

$$\text{Profit ratio} = 14 : 15$$

$$\text{Total ratio units} = 14 + 15 = 29$$

$$\text{Total profit} = \text{Rs. } 87000$$

$$\text{Value of 1 unit} = \frac{87000}{29} = \text{Rs. } 3000$$

$$\text{Bhavna's profit share} = 14 \times 3000 = \text{Rs. } 42000$$

16. Option C

Profit in partnership depends on investment $\times$ time.

Investment ratio of Ishaan and Lakshya = 3 : 5.

Let Ishaan's investment = $3x$ and Lakshya's investment = $5x$.

Let Ishaan invest for t months.

Lakshya invested for 2 months less than Ishaan.

So, Lakshya's time = $t - 2$ months.

Ishaan's profit is 10% less than Lakshya's profit.

So, Ishaan's profit = 90% of Lakshya's profit.

Profit ratio of Ishaan and Lakshya = 90 : 100 = 9 : 10.

Now,

$$3x \times t : 5x \times (t - 2) = 9 : 10$$

$$3t : 5(t - 2) = 9 : 10$$

Using cross multiplication:

$$3t \times 10 = 9 \times 5(t - 2)$$

$$30t = 45(t - 2)$$

$$30t = 45t - 90$$

$$45t - 30t = 90$$

$$15t = 90$$

$$t = 90/15$$

$$t = 6$$

Ishaan invested for 6 months.

Lakshya invested for $t - 2$ months.

Lakshya's time = $6 - 2 = 4$ months.

17. Option E

Initial quantity of milk = 200 liters.

Water added = $(z + 40)$ liters.

Total mixture after adding water = $200 + z + 40 = z + 240$ liters.

Milk in the mixture = 200 liters.

60 liters of the mixture is taken out.

Milk removed from 60 liters mixture = $60 \times 200/(z + 240)$.

After removing 60 liters mixture and adding 60 liters milk, final milk becomes 220 liters.

So,

$$200 - 60 \times 200/(z + 240) + 60 = 220$$

$$260 - 12000/(z + 240) = 220$$

$$12000/(z + 240) = 260 - 220$$

$$12000/(z + 240) = 40$$

$$z + 240 = 12000/40$$

$$z + 240 = 300$$

$$z = 300 - 240$$

$$z = 60$$

18. Option A

Let Nisha's monthly salary be Rs. x .

She spends 25% on house rent and 35% on other expenses.

Total spending = 25% + 35% = 60%.

Remaining salary = 100% - 60% = 40%.

So, remaining amount = 40% of $x = 40x/100$.

This remaining amount is divided between her sister and brother in the ratio 5 : 3.

Difference of ratio parts = 5 - 3 = 2.

Total ratio parts = 5 + 3 = 8.

The difference between the amounts is Rs. 6000.

So, 2 parts = Rs. 6000.

1 part = $6000/2 = \text{Rs. } 3000$.

Total 8 parts = $3000 \times 8 = \text{Rs. } 24000$.

So, 40% of salary = Rs. 24000.

$$40x/100 = 24000$$

$$x = 24000 \times 100/40$$

$$x = 60000$$

19. Option D

Distance between the two cities = 720 km.

The first bus covers 720 km in 6 hours.

Speed of the first bus = $720/6 = 120 \text{ km/hr}$.

The second bus covers 720 km in 12 hours.

Speed of the second bus = $720/12 = 60 \text{ km/hr}$.

Both buses are moving towards each other.

So, relative speed = $120 + 60 = 180 \text{ km/hr}$.

Time taken to meet = Distance/Relative speed.

Time taken = $720/180 = 4 \text{ hours}$.

20. Option B

Use the unit method for quick calculation.

Take total tank capacity = 60 units.

Pipe P fills 20% of the tank in 4 hours.

20% of 60 = 12 units.

So, efficiency of Pipe P = $12/4 = 3$ units per hour.

Pipe Q fills 30% of the tank in 6 hours.

30% of 60 = 18 units.

So, efficiency of Pipe Q = $18/6 = 3$ units per hour.

Pipe R empties the full tank in 30 hours.

So, efficiency of Pipe R = $60/30 = 2$ units per hour.

Since Pipe R is an outlet pipe, its work will be subtracted.

Net efficiency of all three pipes = $3 + 3 - 2 = 4$ units per hour.

The tank is already 40% filled.

So, remaining tank = 60%.

60% of 60 = 36 units.

Time taken to fill the remaining tank = $36/4 = 9$ hours.

Therefore, all three pipes together will fill the remaining tank in 9 hours.

Hence, the correct answer is option (B).

3. Quadratic Equations

CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $4x^2 - 41x + 102 = 0$

II. $5y^2 - 9y - 44 = 0$

2.) I. $6x^2 + 17x - 65 = 0$

II. $7y^2 - 57y + 116 = 0$

3.) I. $6x^2 - 47x + 35 = 0$

II. $2y^2 - 13y + 15 = 0$

4.) I. $5x^2 - 60x + 160 = 0$

II. $7y^2 - 23y - 20 = 0$

5.) I. $7x^2 - 19x - 36 = 0$

II. $2y^2 - 21y + 52 = 0$

6.) I. $4x^2 - 63x + 230 = 0$

II. $6y^2 - 23y - 35 = 0$

7.) I. $5x^2 + 23x - 84 = 0$

II. $6y^2 - 35y + 51 = 0$

8.) I. $8x^2 - 85x + 50 = 0$

II. $6y^2 - 51y + 63 = 0$

9.) I. $4x^2 - 56x + 180 = 0$

II. $7y^2 - 33y - 10 = 0$

10.) I. $8x^2 - 41x - 42 = 0$

II. $3y^2 - 43y + 150 = 0$

11.) I. $7x^2 - 87x + 248 = 0$

II. $5y^2 - 17y - 12 = 0$

12.) I. $9x^2 + 34x - 120 = 0$

II. $5y^2 - 53y + 126 = 0$

13.) I. $4x^2 - 29x + 30 = 0$

II. $8y^2 - 58y + 60 = 0$

14.) I. $3x^2 - 51x + 180 = 0$

II. $7y^2 - 27y - 40 = 0$

15.) I. $5x^2 - 42x + 16 = 0$

II. $2y^2 - 37y + 168 = 0$

16.) I. $3x^2 - 49x + 176 = 0$

II. $8y^2 - 33y - 35 = 0$

17.) I. $8x^2 + 19x - 105 = 0$

II. $5y^2 - 48y + 115 = 0$

18.) I. $9x^2 - 50x - 24 = 0$

II. $2y^2 - 3y - 14 = 0$

19.) I. $5x^2 - 50x + 105 = 0$

II. $4y^2 - y - 33 = 0$

20.) I. $6x^2 - 17x - 65 = 0$

II. $4y^2 - 48y + 140 = 0$

Answers:

1. A

2. B

3. C

4. D

5. E

6. A

7. B

8. C

9. D

10. E

11. A

12. B

13. C

14. D

15. E

16. A

17. B

18. C

19. D

20. E

Solutions:

$$(1) x = 6, \frac{17}{4}$$

$$y = 4, -\frac{11}{5}$$

$$(2) x = \frac{13}{6}, -5$$

$$y = \frac{29}{7}, 4$$

$$(3) x = 7, \frac{5}{6}$$

$$y = 5, \frac{3}{2}$$

$$(4) x = 8, 4$$

$$y = 4, -\frac{5}{7}$$

$$(5) x = 4, -\frac{9}{7}$$

$$y = \frac{13}{2}, 4$$

$$(6) x = 10, \frac{23}{4}$$

$$y = 5, -\frac{7}{6}$$

$$(7) x = \frac{12}{5}, -7$$

$$y = 3, \frac{17}{6}$$

$$(8) x = 10, \frac{5}{8}$$

$$y = 7, \frac{3}{2}$$

$$(9) x = 9, 5$$

$$y = 5, -\frac{2}{7}$$

$$(10) x = 6, -\frac{7}{8}$$

$$y = \frac{25}{3}, 6$$

$$(11) x = 8, \frac{31}{7}$$

$$y = 4, -\frac{3}{5}$$

$$(12) x = \frac{20}{9}, -6$$

$$y = 7, \frac{18}{5}$$

$$(13) x = 6, \frac{5}{4}$$

$$y = 6, \frac{5}{4}$$

$$(14) x = 12, 5$$

$$y = 5, -\frac{8}{7}$$

$$(15) x = 8, \frac{2}{5}$$

$$y = \frac{21}{2}, 8$$

$$(16) x = 11, \frac{16}{3}$$

$$y = 5, -\frac{7}{8}$$

$$(17) x = \frac{21}{8}, -5$$

$$y = 5, \frac{23}{5}$$

$$(18) x = 6, -\frac{4}{9}$$

$$y = \frac{7}{2}, -2$$

$$(19) x = 7, 3$$

$$y = 3, -\frac{11}{4}$$

$$(20) x = 5, -\frac{13}{6}$$

$$y = 7, 5$$

4. WRONG NUMBER SERIES

(1) 520, 540, 570, 618, 668, 740

(a) 570

(b) 668

(c) 618

(d) 740

(e) None of these

(2) 1800, 1778, 1748, 1712, 1682, 1630

(a) 1682

(b) 1778

(c) 1712

(d) 1630

(e) None of these

(3) 134, 257, 759, 2963, 14771, 88571

(a) 257

(b) 2963

(c) 14771

(d) 134

(e) None of these

(4) 95394, 31826, 7970, 1608, 280, 50

(a) 7970

(b) 31826

(c) 1608

(d) 280

(e) None of these

(5) 740, 764, 804, 864, 958, 1060

(a) 804

(b) 764

(c) 1060

(d) 958

(e) None of these

(6) 184, 373, 756, 1517, 3074, 6173

(a) 373

(b) 1517

(c) 756

(d) 6173

(e) None of these

(7) 2200, 2056, 1887, 1681, 1466, 1210

(a) 2056

(b) 1210

(c) 1681

- (d) 1466
(e) None of these

(8) 160, 240, 480, 1220, 3600, 12600

- (a) 1220
(b) 240
(c) 3600
(d) 12600
(e) None of these

(9) 306, 380, 468, 552, 650, 756

- (a) 380
(b) 650
(c) 756
(d) 306
(e) None of these

(10) 500, 518, 1036, 1072, 3226, 3288

- (a) 518
(b) 1036
(c) 3288
(d) 3226
(e) None of these

(11) 245, 259, 301, 427, 805, 1929

- (a) 301
(b) 1929

- (c) 427
(d) 805
(e) None of these

(12) 120, 250, 523, 1110, 2345, 4906

- (a) 523
(b) 2345
(c) 1110
(d) 4906
(e) None of these

(13) 680, 700, 695, 730, 710, 790

- (a) 695
(b) 700
(c) 710
(d) 790
(e) None of these

(14) 70, 424, 2129, 8522, 25621, 51278

- (a) 424
(b) 2129
(c) 8522
(d) 51278
(e) None of these

(15) 925.5, 919, 906, 880, 828, 734

(a) 906

(b) 880

(c) 828

(d) 734

(e) None of these

(16) 165, 475, 1405, 4145, 12385, 37095

(a) 475

(b) 4145

(c) 12385

(d) 37095

(e) None of these

(17) 1040, 1071, 1109, 1154, 1216, 1265

(a) 1216

(b) 1109

(c) 1154

(d) 1265

(e) None of these

(18) 2198, 1114, 598, 348, 238, 200

(a) 598

(b) 348

(c) 238

(d) 1114

(e) None of these

(19) 700, 711, 730, 775, 824, 915

(a) 711

(b) 824

(c) 775

(d) 915

(e) None of these

(20) 450, 470, 970, 2950, 11860, 59310

(a) 970

(b) 11860

(c) 2950

(d) 59310

(e) None of these

Answers:

(1) c

(2) a

(3) e

(4) b

(5) d

(6) b

(7) c

(8) a

(9) e

(10) d

(11) b

(12) e

(13) a

(14) c

(15) d

(16) e

(17) a

(18) d

(19) c

(20) b

Solutions:

(1) +20, +30, +42, +56, +72.

(2) $-(11 \times 2)$, $-(10 \times 3)$, $-(9 \times 4)$, $-(8 \times 5)$, $-(7 \times 6)$.(3) $\times 2 - 11$, $\times 3 - 22$, $\times 4 - 33$, $\times 5 - 44$, $\times 6 - 55$.(4) $\div 3 + 18$, $\div 4 + 16$, $\div 5 + 14$, $\div 6 + 12$, $\div 7 + 10$.(5) $+(3 \times 8)$, $+(4 \times 10)$, $+(5 \times 12)$, $+(6 \times 14)$, $+(7 \times 16)$.(6) $\times 2 + 5$, $\times 2 + 10$, $\times 2 + 15$, $\times 2 + 20$, $\times 2 + 25$.(7) -12^2 , -13^2 , -14^2 , -15^2 , -16^2 .(8) $\times 1.5$, $\times 2$, $\times 2.5$, $\times 3$, $\times 3.5$.

(9) +74, +82, +90, +98, +106.

(10) +18, $\times 2$, +36, $\times 3$, +72.

(11) +14, +42, +126, +378, +1134.

(12) $\times 2 + 2^3$, $\times 2 + 3^3$, $\times 2 + 4^3$, $\times 2 + 5^3$, $\times 2 + 6^3$.

(13) +20, -10, +40, -20, +80.

(14) $\times 6 + 2^2$, $\times 5 + 3^2$, $\times 4 + 4^2$, $\times 3 + 5^2$, $\times 2 + 6^2$.

(15) -6.5, -13, -26, -52, -104.

(16) $\times 3 - 20$, $\times 3 - 30$, $\times 3 - 40$, $\times 3 - 50$, $\times 3 - 60$.

(17) +31, +38, +45, +52, +59.

(18) $\div 2 + 5^2$, $\div 2 + 6^2$, $\div 2 + 7^2$, $\div 2 + 8^2$, $\div 2 + 9^2$.(19) $+(1^2 + 10)$, $+(3^2 + 10)$, $+(5^2 + 10)$, $+(7^2 + 10)$, $+(9^2 + 10)$.(20) $\times 1 + 20$, $\times 2 + 30$, $\times 3 + 40$, $\times 4 + 50$, $\times 5 + 60$.

5. MISSING NUMBER SERIES

(1) 8, 16, 31, ?, 90, 138

- (a) 51
- (b) 53
- (c) 55
- (d) 57
- (e) None of these

(2) 17, 40, 92, ?, 428, 886

- (a) 198
- (b) 200
- (c) 202
- (d) 204
- (e) None of these

(3) 4432, 2220, ?, 190, 42, 11

- (a) 742
- (b) 744
- (c) 746
- (d) 748
- (e) None of these

(4) 6, 15, 39, ?, 291, 825

- (a) 105
- (b) 103
- (c) 107
- (d) 109
- (e) None of these

(5) 22, 32, ?, 108, 190, 312

(a) 54

(b) 56

(c) 58

(d) 60

(e) None of these

(6) 800, 788, 761, ?, 638, 530

(a) 711

(b) 713

(c) 715

(d) 717

(e) None of these

(7) 2380, 2361, 2324, ?, 2196, 2105

(a) 2263

(b) 2265

(c) 2267

(d) 2269

(e) None of these

(8) 62.4, 66, 71.4, ?, 87.6, 98.4

(a) 78.6

(b) 77.6

(c) 79.6

(d) 80.6

(e) None of these

(9) 58, 79, 111, ?, 216, 293

(a) 154

(b) 156

(c) 158

- (d) 160
(e) None of these

(10) 16, 45, 96, ?, 288, 441

- (a) 171
(b) 173
(c) 175
(d) 177
(e) None of these

(11) 128, 267, 552, ?, 2290, 4619

- (a) 1125
(b) 1127
(c) 1129
(d) 1131
(e) None of these

(12) 7560, 1080, 180, ?, 9, 3

- (a) 34
(b) 36
(c) 38
(d) 40
(e) None of these

(13) 125, 143, 170, 209, ?, 335

- (a) 257
(b) 259
(c) 261
(d) 263
(e) None of these

(14) 73, 144, 429, ?, 8555, 51324

- (a) 1712
(b) 1714
(c) 1716
(d) 1718
(e) None of these

(15) 7, 15, 38, ?, 270, 665

- (a) 99
(b) 101
(c) 103
(d) 105
(e) None of these

(16) 2800, 2788, 2752, ?, 2522, 2270

- (a) 2666
(b) 2668
(c) 2670
(d) 2672
(e) None of these

(17) 146, 169, 215, ?, 376, 491

- (a) 282
(b) 284
(c) 286
(d) 288
(e) None of these

(18) 87, 97, 127, ?, 287, 437

- (a) 183
(b) 185
(c) 187
(d) 189
(e) None of these

(19) 2200, 2225, 2275, ?, 2575, 2975

- (a) 2375
(b) 2370
(c) 2380
(d) 2385
(e) None of these

(20) 14, 78, 252, ?, 1290, 2394

- (a) 618
(b) 620
(c) 622
(d) 624
(e) None of these

Answers:

- (1) c
- (2) c
- (3) b
- (4) a
- (5) c
- (6) b
- (7) d
- (8) a
- (9) b
- (10) c
- (11) c
- (12) b
- (13) d
- (14) a
- (15) c
- (16) d
- (17) b
- (18) c
- (19) a
- (20) b

Solutions:

- (1) +8, +15, +24, +35, +48
Pattern: add $n^2 + 2n$, for $n = 2, 3, 4, 5, 6$
- (2) $\times 2 + 6$, $\times 2 + 12$, $\times 2 + 18$, $\times 2 + 24$,
 $\times 2 + 30$
- (3) $\div 2 + 4$, $\div 3 + 4$, $\div 4 + 4$, $\div 5 + 4$, $\div 6 + 4$
- (4) $\times 2 + 3$, $\times 2 + 9$, $\times 2 + 27$, $\times 2 + 81$, $\times 2 + 243$

$$(5) +3^2 + 1, +5^2 + 1, +7^2 + 1, +9^2 + 1, +11^2 + 1$$

$$(6) -3 \times 2^2, -3 \times 3^2, -3 \times 4^2, -3 \times 5^2, -3 \times 6^2$$

$$(7) -19, -37, -55, -73, -91$$

$$(8) +3.6, +5.4, +7.2, +9, +10.8$$

$$(9) +3 \times 7, +4 \times 8, +5 \times 9, +6 \times 10, +7 \times 11$$

$$(10) \text{ Terms are: } n^3 + 2n^2, \text{ for } n = 2, 3, 4, 5, 6, 7$$

$$(11) \times 2 + 11, \times 2 + 18, \times 2 + 25, \times 2 + 32, \times 2 + 39$$

$$(12) \div 7, \div 6, \div 5, \div 4, \div 3$$

$$(13) +18, +27, +39, +54, +72$$

$$\text{Second difference: } +9, +12, +15, +18$$

$$(14) \times 2 - 2, \times 3 - 3, \times 4 - 4, \times 5 - 5, \times 6 - 6$$

$$(15) \times 2 + 1^3, \times 2 + 2^3, \times 2 + 3^3, \times 2 + 4^3, \times 2 + 5^3$$

$$(16) -(2^3 + 2^2), -(3^3 + 3^2), -(4^3 + 4^2), -(5^3 + 5^2), -(6^3 + 6^2)$$

$$(17) +23 \times 1, +23 \times 2, +23 \times 3, +23 \times 4, +23 \times 5$$

$$(18) +10, +30, +60, +100, +150$$

$$\text{Second difference: } +20, +30, +40, +50$$

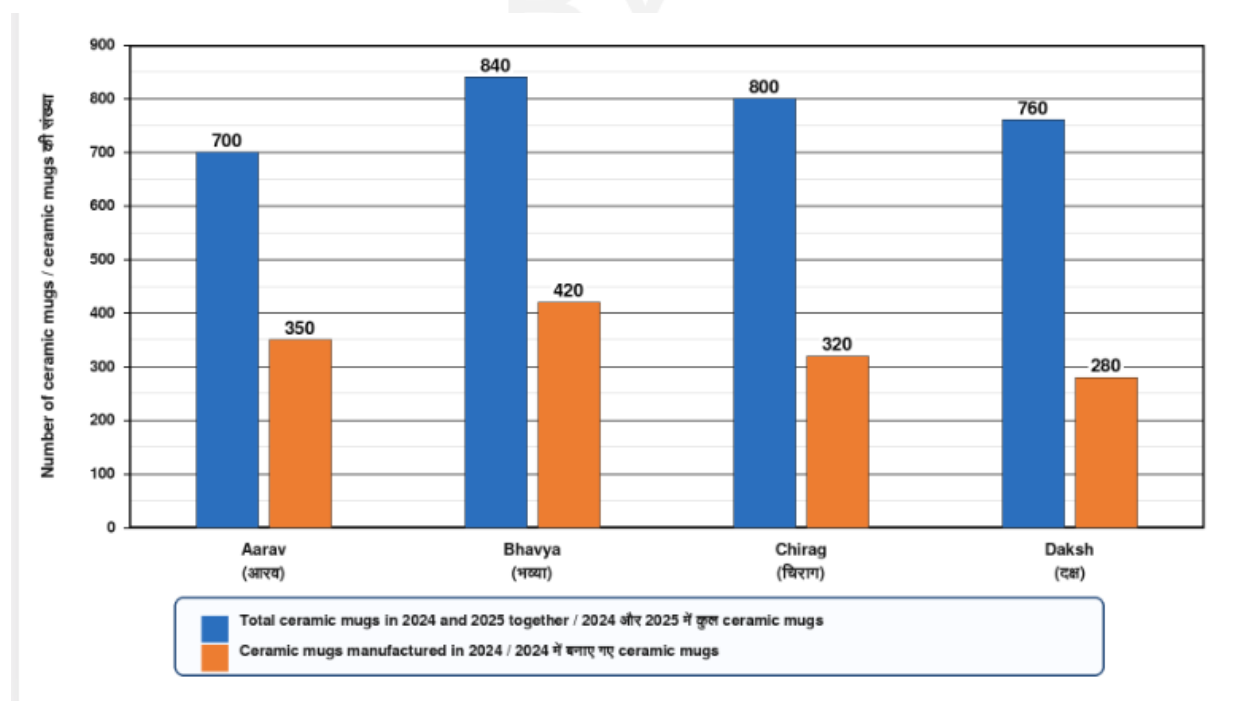
$$(19) +25, +50, +100, +200, +400$$

$$(20) \text{ Terms are: } n^4 - n, \text{ for } n = 2, 3, 4, 5, 6, 7$$

6. DATA INTERPRETATION

Set 1: The bar graph shows the number of ceramic mugs manufactured by four pottery studios in two different years. The first bar shows the total number of ceramic mugs manufactured in 2024 and 2025 together, and the second bar shows the number of ceramic mugs manufactured in 2024.

दंड आलेख चार मिट्टी के बर्तन स्टूडियो द्वारा दो अलग-अलग वर्षों में बनाए गए ceramic mugs की संख्या दर्शाता है। पहला बार 2024 और 2025 में मिलाकर बनाए गए कुल ceramic mugs को दर्शाता है, और दूसरा बार 2024 में बनाए गए ceramic mugs को दर्शाता है।



Q1: The number of ceramic mugs manufactured in 2025 by Aarav is what percentage more or less than the number of ceramic mugs manufactured in 2024 by Daksh?

आरव द्वारा 2025 में बनाए गए ceramic mugs, दक्ष द्वारा 2024 में बनाए गए ceramic mugs से कितने प्रतिशत अधिक या कम हैं?

(A) 15%

- (B) 20%
- (C) 25%
- (D) 30%
- (E) 35%

Q2: The number of ceramic mugs manufactured in 2026 by Aarav is $33\frac{1}{3}\%$ more than the number of ceramic mugs manufactured in 2025 by Bhavya. If the average number of ceramic mugs manufactured in 2026 by Aarav and Bhavya is 630, then find the number of ceramic mugs manufactured in 2026 by Bhavya.

आरव द्वारा 2026 में बनाए गए ceramic mugs, भव्या द्वारा 2025 में बनाए गए ceramic mugs से $33\frac{1}{3}\%$ अधिक हैं। यदि 2026 में आरव और भव्या द्वारा बनाए गए ceramic mugs की औसत संख्या 630 है, तो भव्या द्वारा 2026 में बनाए गए ceramic mugs की संख्या ज्ञात कीजिए।

- (A) 640
- (B) 670
- (C) 680
- (D) 720
- (E) 700

Q3: The number of painted mugs manufactured in 2025 by Chirag is $\frac{5}{7}$ of the number of ceramic mugs manufactured in 2024 by Bhavya. Find the difference between the number of painted mugs manufactured in 2025 by Chirag and the number of ceramic mugs manufactured in 2024 by Daksh.

चिराग द्वारा 2025 में बनाए गए painted mugs, भव्या द्वारा 2024 में बनाए गए ceramic mugs के $\frac{5}{7}$ हैं। चिराग द्वारा 2025 में बनाए गए painted mugs और दक्ष द्वारा 2024 में बनाए गए ceramic mugs की संख्या का अंतर ज्ञात कीजिए।

- (A) 20
- (B) 40
- (C) 60
- (D) 80
- (E) 100

Q4: Find the average number of ceramic mugs manufactured in 2025 by Chirag and Daksh.

चिराग और दक्ष द्वारा 2025 में बनाए गए ceramic mugs की औसत संख्या ज्ञात कीजिए।

- (A) 420
- (B) 450
- (C) 460
- (D) 480
- (E) 520

Q5: The total number of decorative mugs manufactured in 2026 by a new pottery studio Eshan is $12\frac{1}{2}\%$ more than the number of ceramic mugs manufactured in 2025 by Daksh. If the number of rejected decorative mugs in Eshan is $37\frac{1}{2}\%$ of the number of ceramic mugs manufactured in 2024 by Chirag, then find the number of saleable decorative mugs manufactured by Eshan in 2026.

एक नए मिट्टी के बर्तन स्टूडियो ईशान द्वारा 2026 में बनाए गए कुल decorative mugs, दक्ष द्वारा 2025 में बनाए गए ceramic mugs से $12\frac{1}{2}\%$ अधिक हैं। यदि ईशान में rejected decorative mugs की संख्या, चिराग द्वारा 2024 में बनाए गए ceramic mugs की $37\frac{1}{2}\%$ है, तो 2026 में ईशान द्वारा बनाए गए saleable decorative mugs की संख्या ज्ञात कीजिए।

- (A) 390
- (B) 420
- (C) 450
- (D) 470
- (E) 500

Solutions:

Pottery Studio स्टूडियो	Total mugs (2024 + 2025) कुल mugs	Mugs in 2024 2024	Mugs in 2025 2025
Aarav (आरव)	700	350	350
Bhavya (भव्या)	840	420	420
Chirag (चिराग)	800	320	480
Daksh (दक्ष)	760	280	480
Total / कुल	3100	1370	1730

1. Option C

Aarav total in 2024 and 2025 = 700

Aarav in 2024 = 350

Aarav in 2025 = 700 – 350 = 350

Daksh in 2024 = 280

Percentage more = $(350 - 280) / 280 \times 100 = 25\%$

2. Option E

Bhavya total in 2024 and 2025 = 840

Bhavya in 2024 = 420

Bhavya in 2025 = 840 – 420 = 420

Aarav in 2026 = $420 \times \frac{4}{3} = 560$

Average of Aarav and Bhavya in 2026 = 630

Total of Aarav and Bhavya in 2026 = $630 \times 2 = 1260$

Bhavya in 2026 = 1260 – 560 = 700

3. Option A

Bhavya in 2024 = 420

Painted mugs by Chirag in 2025 = $420 \times \frac{5}{7} = 300$

Daksh in 2024 = 280

Difference = $300 - 280 = 20$

4. Option D

Chirag in 2025 = $800 - 320 = 480$

Daksh in 2025 = $760 - 280 = 480$

Average = $(480 + 480) / 2 = 480$

5. Option B

Daksh in 2025 = $760 - 280 = 480$

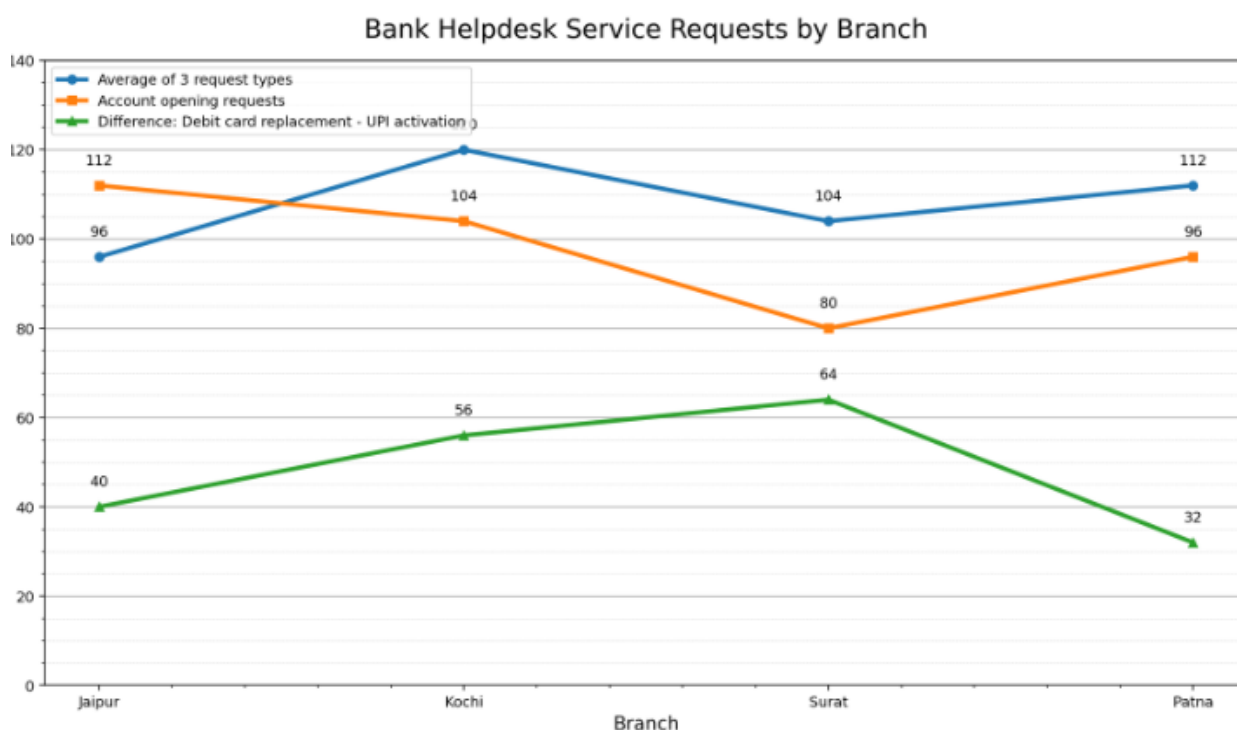
Total decorative mugs by Eshan in 2026 = $480 \times \frac{9}{8} = 540$

Rejected decorative mugs = $320 \times \frac{3}{8} = 120$

Saleable decorative mugs = $540 - 120 = 420$

Set 2: The line graph shows the average number of three types of banking service requests, i.e., account opening requests, debit card replacement requests and UPI activation requests handled by a bank helpdesk in four different branches. The line graph also shows the number of account opening requests and the difference between debit card replacement requests and UPI activation requests. In each branch, the number of debit card replacement requests is more than the number of UPI activation requests. Study the line graph carefully and answer the following questions.

रेखा आलेख चार अलग-अलग शाखाओं में बैंक हेल्पडेस्क द्वारा संभाले गए तीन प्रकार के बैंकिंग सेवा अनुरोधों, अर्थात् खाता खोलने के अनुरोध, डेबिट कार्ड बदलने के अनुरोध और UPI सक्रियण अनुरोधों की औसत संख्या को दर्शाता है। रेखा आलेख खाता खोलने के अनुरोधों की संख्या और डेबिट कार्ड बदलने के अनुरोधों तथा UPI सक्रियण अनुरोधों के बीच अंतर भी दर्शाता है। प्रत्येक शाखा में डेबिट कार्ड बदलने के अनुरोधों की संख्या UPI सक्रियण अनुरोधों से अधिक है। रेखा आलेख का ध्यानपूर्वक अध्ययन करें और निम्नलिखित प्रश्नों के उत्तर दें।



Q1: The total number of requests handled by the Bhopal branch is 25% less than the total number of requests handled by the Surat branch. The number of UPI activation requests handled by Bhopal is the same as that handled by Surat. If the ratio of account opening requests to debit card replacement requests handled by

Bhopal is 7 : 8, then find the number of account opening requests handled by Bhopal.

भोपाल शाखा द्वारा संभाले गए कुल अनुरोध, सूरत शाखा द्वारा संभाले गए कुल अनुरोधों से 25% कम हैं। भोपाल द्वारा संभाले गए UPI सक्रियण अनुरोधों की संख्या सूरत द्वारा संभाले गए UPI सक्रियण अनुरोधों के बराबर है। यदि भोपाल द्वारा संभाले गए खाता खोलने के अनुरोधों और डेबिट कार्ड बदलने के अनुरोधों का अनुपात 7 : 8 है, तो भोपाल द्वारा संभाले गए खाता खोलने के अनुरोधों की संख्या ज्ञात कीजिए।

- (A) 64
- (B) 84
- (C) 72
- (D) 70
- (E) 76

Q2: If the service fee for each account opening request, debit card replacement request and UPI activation request handled by the Surat branch is Rs. 25, Rs. 15 and Rs. A respectively, and the total service fee earned by the Surat branch from these three services is Rs. 5,900, then find the value of A.

यदि सूरत शाखा द्वारा संभाले गए प्रत्येक खाता खोलने के अनुरोध, डेबिट कार्ड बदलने के अनुरोध और UPI सक्रियण अनुरोध का सेवा शुल्क क्रमशः Rs. 25, Rs. 15 और Rs. A है, और इन तीन सेवाओं से सूरत शाखा द्वारा अर्जित कुल सेवा शुल्क Rs. 5,900 है, तो A का मान ज्ञात कीजिए।

- (A) 18
- (B) 20
- (C) 24
- (D) 15
- (E) 30

Q3: Find the number of debit card replacement requests handled by the Jaipur branch as a percentage of the total number of account opening requests and UPI activation requests handled by the same branch.

जयपुर शाखा द्वारा संभाले गए डेबिट कार्ड बदलने के अनुरोधों की संख्या, उसी शाखा द्वारा संभाले गए खाता खोलने के अनुरोधों और UPI सक्रियण अनुरोधों की कुल संख्या का कितना प्रतिशत है?

- (A) 50%

- (B) 56%
- (C) 62.5%
- (D) $66\frac{2}{3}\%$
- (E) 60%

Q4: If the average number of UPI activation requests handled by Jaipur, Kochi, Surat, Patna and Indore branches together is 92, then find the number of UPI activation requests handled by the Indore branch.

यदि जयपुर, कोच्चि, सूरत, पटना और इंदौर शाखाओं द्वारा मिलकर संभाले गए UPI सक्रियण अनुरोधों की औसत संख्या 92 है, तो इंदौर शाखा द्वारा संभाले गए UPI सक्रियण अनुरोधों की संख्या ज्ञात कीजिए।

- (A) 104
- (B) 96
- (C) 112
- (D) 100
- (E) 108

Q5: In the Ranchi branch, the number of account opening requests and debit card replacement requests increased by 12.5% and 25% respectively as compared to the Kochi branch, while the total number of requests handled by Ranchi remained the same as that handled by Kochi. Find the percentage decrease in UPI activation requests handled by Ranchi as compared to Kochi.

रांची शाखा में खाता खोलने के अनुरोधों और डेबिट कार्ड बदलने के अनुरोधों की संख्या कोच्चि शाखा की तुलना में क्रमशः 12.5% और 25% बढ़ गई, जबकि रांची द्वारा संभाले गए कुल अनुरोधों की संख्या कोच्चि द्वारा संभाले गए कुल अनुरोधों के बराबर रही। कोच्चि की तुलना में रांची द्वारा संभाले गए UPI सक्रियण अनुरोधों में प्रतिशत कमी ज्ञात कीजिए।

- (A) 48%
- (B) 45%
- (C) 52%
- (D) 50%
- (E) 56%

Solutions:

On Jaipur branch

Total requests = Average $\times$ 3 = $96 \times 3 = 288$

Account opening requests = 112

Debit card replacement requests + UPI activation requests = $288 - 112 = 176$

Debit card replacement requests - UPI activation requests = 40

Debit card replacement requests = $(176 + 40) / 2 = 108$

UPI activation requests = $(176 - 40) / 2 = 68$

On Kochi branch

Total requests = Average $\times$ 3 = $120 \times 3 = 360$

Account opening requests = 104

Debit card replacement requests + UPI activation requests = $360 - 104 = 256$

Debit card replacement requests - UPI activation requests = 56

Debit card replacement requests = $(256 + 56) / 2 = 156$

UPI activation requests = $(256 - 56) / 2 = 100$

On Surat branch

Total requests = Average $\times$ 3 = $104 \times 3 = 312$

Account opening requests = 80

Debit card replacement requests + UPI activation requests = $312 - 80 = 232$

Debit card replacement requests - UPI activation requests = 64

Debit card replacement requests = $(232 + 64) / 2 = 148$

UPI activation requests = $(232 - 64) / 2 = 84$

On Patna branch

Total requests = Average $\times$ 3 = $112 \times 3 = 336$

Account opening requests = 96

Debit card replacement requests + UPI activation requests = $336 - 96 = 240$

Debit card replacement requests - UPI activation requests = 32

Debit card replacement requests = $(240 + 32) / 2 = 136$

UPI activation requests = $(240 - 32) / 2 = 104$

Branch	Average	Account Opening	Difference (Debit-UPI)	Total	Debit Card Replacement	UPI Activation
Jaipur	96	112	40	288	108	68
Kochi	120	104	56	360	156	100
Surat	104	80	64	312	148	84
Patna	112	96	32	336	136	104

1. Option D

Total requests in Surat = $104 \times 3 = 312$

Total requests in Bhopal = $312 \times 75\% = 234$

In Surat, account opening requests = 80

Debit card replacement requests + UPI activation requests = $312 - 80 = 232$

Debit card replacement requests - UPI activation requests = 64

UPI activation requests in Surat = $(232 - 64) \div 2 = 84$

UPI activation requests in Bhopal = 84

Account opening requests + debit card replacement requests in Bhopal = $234 - 84 = 150$

Required account opening requests = $150 \times 7 \div 15 = 70$

2. Option B

Total requests in Surat = $104 \times 3 = 312$

Account opening requests in Surat = 80

Debit card replacement requests + UPI activation requests = $312 - 80 = 232$

Debit card replacement requests - UPI activation requests = 64

Debit card replacement requests = $(232 + 64) \div 2 = 148$

UPI activation requests = $(232 - 64) \div 2 = 84$

Total fee = $80 \times 25 + 148 \times 15 + 84 \times A$

$5900 = 2000 + 2220 + 84A$

$84A = 1680$

$A = 20$

3. Option E

Total requests in Jaipur = $96 \times 3 = 288$

Account opening requests in Jaipur = 112

Debit card replacement requests + UPI activation requests = $288 - 112 = 176$

Debit card replacement requests - UPI activation requests = 40

Debit card replacement requests = $(176 + 40) \div 2 = 108$

UPI activation requests = $(176 - 40) \div 2 = 68$

Required percentage = $108 \times 100 \div (112 + 68)$

Required percentage = $108 \times 100 \div 180 = 60\%$

4. Option A

UPI activation requests in Jaipur = 68

UPI activation requests in Kochi = $[(120 \times 3 - 104) - 56] \div 2 = 100$

UPI activation requests in Surat = 84

UPI activation requests in Patna = $[(112 \times 3 - 96) - 32] \div 2 = 104$

Total UPI activation requests in four branches = $68 + 100 + 84 + 104 = 356$

Total UPI activation requests in five branches = $92 \times 5 = 460$

UPI activation requests in Indore = $460 - 356 = 104$

5. Option C

Total requests in Kochi = $120 \times 3 = 360$

Account opening requests in Kochi = 104

Debit card replacement requests + UPI activation requests = $360 - 104 = 256$

Debit card replacement requests - UPI activation requests = 56

Debit card replacement requests in Kochi = $(256 + 56) \div 2 = 156$

UPI activation requests in Kochi = $(256 - 56) \div 2 = 100$

Account opening requests in Ranchi = $104 \times 112.5\% = 104 \times 9/8 = 117$

Debit card replacement requests in Ranchi = $156 \times 125\% = 195$

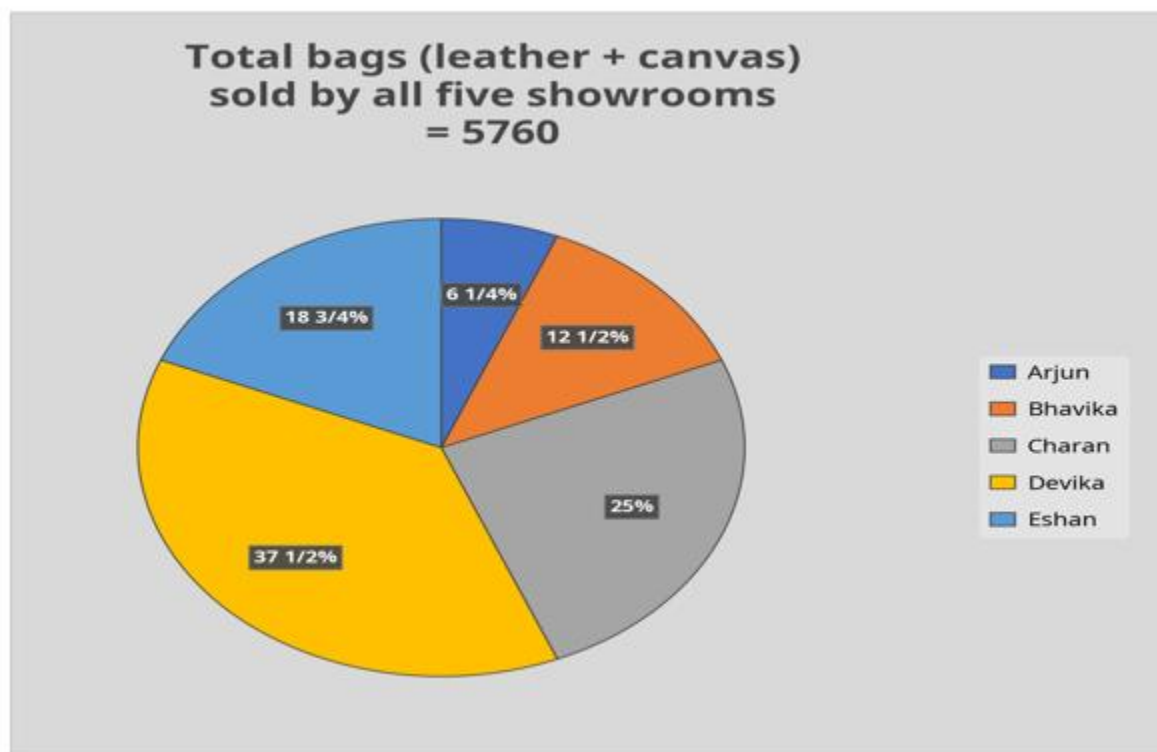
UPI activation requests in Ranchi = $360 - 117 - 195 = 48$

Decrease = $100 - 48 = 52$

Required percentage decrease = $52 \times 100 \div 100 = 52\%$

Set 3: The pie chart shows the percentage distribution of the total number of bags, leather and canvas together, sold by five different showrooms: Arjun, Bhavika, Charan, Devika and Eshan. The total number of bags sold by all five showrooms together is 5760. Read the pie chart carefully and answer the following questions.

पाई चार्ट पाँच अलग-अलग शोरूम अर्जुन, भाविका, चरण, देविका और ईशान द्वारा बेचे गए कुल बैग, यानी चमड़े और कैनवास बैग मिलाकर, का प्रतिशत वितरण दर्शाता है। सभी पाँचों शोरूम द्वारा बेचे गए कुल बैग की संख्या 5760 है। पाई चार्ट को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: Find the difference between the total number of bags sold by Devika and the total number of bags sold by Bhavika.

देविका द्वारा बेचे गए कुल बैग और भाविका द्वारा बेचे गए कुल बैग के बीच का अंतर ज्ञात कीजिए।

- (A) 1260
- (B) 1530
- (C) 1350
- (D) 1440
- (E) 1620

Q2: If the number of leather bags sold by Eshan is 50% more than the number of canvas bags sold by the same showroom, then the number of canvas bags sold by Eshan is what percentage of the total number of bags sold by Arjun?

यदि ईशान द्वारा बेचे गए चमड़े के बैग, उसी शोरूम द्वारा बेचे गए कैनवास बैग से 50% अधिक हैं, तो ईशान द्वारा बेचे गए कैनवास बैग, अर्जुन द्वारा बेचे गए कुल बैग का कितना प्रतिशत हैं?

- (A) 112%
- (B) 120%
- (C) 125%
- (D) 115%
- (E) 108%

Q3: Find the ratio of the total number of bags sold by Charan to the total number of bags sold by Arjun and Bhavika together.

चरण द्वारा बेचे गए कुल बैग का अर्जुन और भाविका द्वारा मिलाकर बेचे गए कुल बैग से अनुपात ज्ञात कीजिए।

- (A) 5:3
- (B) 3:2
- (C) 7:5
- (D) 5:4
- (E) 4:3

Q4: If the total number of bags sold by a new showroom Karan is 324 more than that sold by Devika, then find the average number of bags sold by Karan and Eshan.

यदि एक नए शोरूम करण द्वारा बेचे गए कुल बैग, देविका द्वारा बेचे गए कुल बैग से 324 अधिक हैं, तो करण और ईशान द्वारा बेचे गए बैग की औसत संख्या ज्ञात कीजिए।

- (A) 1782
- (B) 1860
- (C) 1692
- (D) 1728
- (E) 1818

Q5: Find the corresponding central angle, in degrees, for the total number of bags sold by Bhavika and Charan together.

भाविका और चरण द्वारा मिलाकर बेचे गए कुल बैग के लिए संबंधित केंद्रीय कोण, डिग्री में, ज्ञात कीजिए।

- (A) 117
(B) 126
(C) 135
(D) 162
(E) 153

Solutions:

Showroom	Percentage	Fraction Form	Calculation	Pie-chart Value	Actual Bags Sold
Arjun	6 1/4%	1/16	$720 \times 1/16$	45	360
Bhavika	12 1/2%	1/8	$720 \times 1/8$	90	720
Charan	25%	1/4	$720 \times 1/4$	180	1440
Devika	37 1/2%	3/8	$720 \times 3/8$	270	2160
Eshan	18 3/4%	3/16	$720 \times 3/16$	135	1080
Total	100%	1	720×1	720	5760

Total bags sold = 5,760 | Total pie-chart value = 720 | Multiplier = $5,760/720 = 8$

Showroom	Pie-chart Value	Percentage Distribution	Calculation	Total Bags Sold
Arjun	45	6 1/4%	$45 \times 8 = 360$	360
Bhavika	90	12 1/2%	$90 \times 8 = 720$	720
Charan	180	25%	$180 \times 8 = 1440$	1,440
Devika	270	37 1/2%	$270 \times 8 = 2160$	2,160
Eshan	135	18 3/4%	$135 \times 8 = 1080$	1,080
Total	720	100%	720×8	5,760

Common Calculation:

Total pie-chart value = $45 + 90 + 180 + 270 + 135 = 720$

Total bags sold = 5760

Arjun = $6\frac{1}{4}\%$ of 5760 = 360

Bhavika = $12\frac{1}{2}\%$ of 5760 = 720

Charan = 25% of 5760 = 1440

Devika = $37\frac{1}{2}\%$ of 5760 = 2160

Eshan = $18\frac{3}{4}\%$ of 5760 = 1080

1. Option D

Devika sold = 2160

Bhavika sold = 720

Required difference = $2160 - 720 = 1440$

2. Option B

Let canvas bags sold by Eshan = x

Leather bags sold by Eshan = 150% of $x = 1.5x$

Total bags sold by Eshan = $x + 1.5x = 2.5x$

$2.5x = 1080$

$x = 432$

Required percentage = $432/360 \times 100 = 120\%$

3. Option E

Charan sold = 1440

Arjun + Bhavika = $360 + 720 = 1080$

Required ratio = $1440:1080 = 4:3$

4. Option A

Devika sold = 2160

Karan sold = $2160 + 324 = 2484$

Eshan sold = 1080

Required average = $(2484 + 1080)/2 = 1782$

5. Option C

Bhavika + Charan = $12\frac{1}{2}\% + 25\% = 37\frac{1}{2}\%$

Central angle = $37\frac{1}{2}\%$ of 360

Central angle = 135 degrees

CHECKLIST
BY
AASHISH
ARORA

Set 4: The data is about students enrolled in two music courses, Vocal and Guitar, at two music academies, Swarangan Academy and Layamitra Academy. Each student enrolled either only in Vocal, only in Guitar, or in both Vocal and Guitar. The number of students enrolled only in Guitar at Layamitra Academy is 156. The number of students enrolled only in Vocal at Swarangan Academy and Layamitra Academy is in the ratio 2 : 3, respectively. The number of students enrolled in both Vocal and Guitar at Swarangan Academy is 40% of the number of students enrolled only in Vocal at Swarangan Academy. The number of students enrolled in both Vocal and Guitar at Layamitra Academy is 39 more than that at Swarangan Academy. The total number of students enrolled in Vocal at Layamitra Academy is 286. The number of students enrolled only in Guitar at Swarangan Academy is 65 more than that at Layamitra Academy.

यह डेटा दो संगीत अकादमियों, स्वरांगन अकादमी और लयमित्र अकादमी, में दो संगीत कोर्स, गायन और गिटार, में नामांकित छात्रों की संख्या के बारे में है। प्रत्येक छात्र ने या तो केवल गायन, केवल गिटार, या गायन और गिटार दोनों में नामांकन किया। लयमित्र अकादमी में केवल गिटार में नामांकित छात्रों की संख्या 156 है। स्वरांगन अकादमी और लयमित्र अकादमी में केवल गायन में नामांकित छात्रों की संख्या का अनुपात क्रमशः 2 : 3 है। स्वरांगन अकादमी में गायन और गिटार दोनों में नामांकित छात्रों की संख्या, स्वरांगन अकादमी में केवल गायन में नामांकित छात्रों की संख्या का 40% है। लयमित्र अकादमी में गायन और गिटार दोनों में नामांकित छात्रों की संख्या, स्वरांगन अकादमी की तुलना में 39 अधिक है। लयमित्र अकादमी में गायन में नामांकित कुल छात्रों की संख्या 286 है। स्वरांगन अकादमी में केवल गिटार में नामांकित छात्रों की संख्या, लयमित्र अकादमी की तुलना में 65 अधिक है।

Q1: If the simplified ratio of students enrolled only in Guitar at Swarangan Academy to those enrolled only in Guitar at Layamitra Academy is $a : b$, find the value of $a + b$.

यदि स्वरांगन अकादमी में केवल गिटार में नामांकित छात्रों और लयमित्र अकादमी में केवल गिटार में नामांकित छात्रों का सरलतम अनुपात $a : b$ है, तो $a + b$ का मान ज्ञात कीजिए।

- (A) 25
- (B) 27
- (C) 31
- (D) 29
- (E) 33

Q2: The number of students enrolled in both Vocal and Guitar at Layamitra Academy is what percentage more than that at Swarangan Academy?

लयमित्र अकादमी में गायन और गिटार दोनों में नामांकित छात्रों की संख्या, स्वरांगन अकादमी की तुलना में कितने प्रतिशत अधिक है?

- (A) 65%
- (B) 75%
- (C) 70%
- (D) 80%
- (E) 85%

Q3: Find the difference between the number of students enrolled only in Vocal at Swarangan Academy and the number of students enrolled in both Vocal and Guitar at Layamitra Academy.

स्वरांगन अकादमी में केवल गायन में नामांकित छात्रों और लयमित्र अकादमी में गायन और गिटार दोनों में नामांकित छात्रों की संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 52
- (B) 26
- (C) 65
- (D) 13
- (E) 39

Q4: Find the total number of students enrolled in Guitar at Swarangan Academy.

स्वरांगन अकादमी में गिटार में नामांकित कुल छात्रों की संख्या ज्ञात कीजिए।

- (A) 273
- (B) 260
- (C) 286
- (D) 299
- (E) 247

Q5: If the simplified ratio of students enrolled in both Vocal and Guitar at Swarangan Academy to students enrolled only in Guitar at Layamitra Academy is $x : y$, find the value of $x + y$.

यदि स्वरांगन अकादमी में गायन और गिटार दोनों में नामांकित छात्रों और लयमित्र अकादमी में केवल गिटार में नामांकित छात्रों का सरलतम अनुपात $x : y$ है, तो $x + y$ का मान ज्ञात

कीजिए।

- (A) 6
- (B) 5
- (C) 4
- (D) 7
- (E) 8

Solutions:

Category	Swarangan Academy	Layamitra Academy	Total
Only Vocal	130	195	325
Only Guitar	221	156	377
Both Vocal and Guitar	52	91	143
Total	403	442	845

1. Option D

Only Guitar at Swarangan Academy = 221

Only Guitar at Layamitra Academy = 156

Required ratio = 221 : 156

= 17 : 12

a + b = 17 + 12

= 29

2. Option B

Both Vocal and Guitar at Layamitra Academy = 91

Both Vocal and Guitar at Swarangan Academy = 52

Percentage more = $(91 - 52) \div 52 \times 100$

= $39 \div 52 \times 100$

= 75%

3. Option E

Only Vocal at Swarangan Academy = 130

Both Vocal and Guitar at Layamitra Academy = 91

Required difference = $130 - 91$
= 39

4. Option A

Only Guitar at Swarangan Academy = 221

Both Vocal and Guitar at Swarangan Academy = 52

Total students enrolled in Guitar at Swarangan Academy = $221 + 52$
= 273

5. Option C

Both Vocal and Guitar at Swarangan Academy = 52

Only Guitar at Layamitra Academy = 156

Required ratio = 52 : 156

= 1 : 3

$x + y = 1 + 3$

= 4